

Module 7: AI Ethics & Safety - Complete Guide

Understand AI bias, responsible AI development, privacy concerns, and AI regulation.

7.1 What is AI Ethics?

AI ethics is a set of moral principles to ensure AI technologies are developed and used responsibly.

- Ensuring AI systems are fair and unbiased
- Protecting privacy and personal data
- Maintaining transparency and explainability
- Ensuring accountability for AI decisions
- Preventing misuse of AI technology

7.2 AI Bias & Fairness

AI systems can inherit biases from training data, leading to unfair outcomes.

- Types of bias: Data bias, Algorithmic bias, Selection bias
- Examples: Hiring algorithms favoring certain demographics
- Facial recognition performing worse on certain groups
- Credit scoring algorithms with racial bias
- Solutions: Diverse training data, bias audits, fairness metrics

7.3 AI Privacy Concerns

- Data collection and consent
- Surveillance and facial recognition
- Deepfakes and misinformation
- Data security and breaches
- Right to explanation (GDPR)
- Data minimization principles

7.4 Responsible AI Development

- Conduct impact assessments before deployment

- Ensure diverse and representative training data
- Implement transparency and explainability
- Establish human oversight and control
- Regular auditing and monitoring
- Clear accountability frameworks

7.5 Future of AI Regulation

- EU AI Act: Comprehensive AI regulation framework
- US AI Bill of Rights: Guidelines for AI development
- China AI Regulations: Rules for generative AI
- UK AI Safety Institute: AI safety research
- Global AI governance initiatives